Vehicle Load Modeling Based on Real-World Driving Profiles and Dynamics

Physics-based vehicle dynamics | Least-squares identification | Random Forest residual learning

Project overview

Accurate battery-energy prediction is challenging under real-world driving conditions because measured power demand depends on road slope, vehicle dynamics, powertrain losses and operating effects that are not fully observable. This project developed a hybrid approach that retains a longitudinal vehicle-dynamics baseline while using Random Forest regression to correct its remaining battery-power error.

Tesla Model 3 trip recordings were preprocessed and combined with digital elevation data to reconstruct road slope. Vehicle parameters were identified through least-squares optimization, and a residual learner was evaluated on held-out driving trips.

Headline findings — held-out test trips

Dataset	Baseline mean absolute trip-energy error	Hybrid error	Trips improved
404 recorded trips; 62 test trips	12.59%	7.86%	53 of 62 (85.5%)

Interpretation: The test-set mean absolute trip-energy percentage error decreased by **4.73 percentage points, a 37.6% relative reduction**. This is a modelling-error reduction, not a measured improvement in vehicle energy efficiency.

Project scope

The framework estimates battery power and trip energy from recorded driving signals. It is not a battery electrochemistry model, a production vehicle controller or a multi-vehicle validation study. The publicly released portfolio excludes raw trips, precise route coordinates and original source code.

02 / REAL-WORLD DRIVING DATA

The study used 404 TeslaMate-recorded trips from one Tesla Model 3. Available signals included timestamps, speed, battery power, state of charge, outside temperature, elevation and position. Recorded data required cleaning before they could be used consistently for physics-based estimation and residual learning.

Preprocessing strategy

Trip signals were screened for irregular sampling and gaps, smoothed to suppress measurement noise, and standardized to a 1 Hz working frequency. Acceleration was derived from the cleaned speed signal, with attention to the sensitivity of numerical differentiation to noise. Each processed trip preserved temporal alignment among speed, battery power and reconstructed terrain inputs.

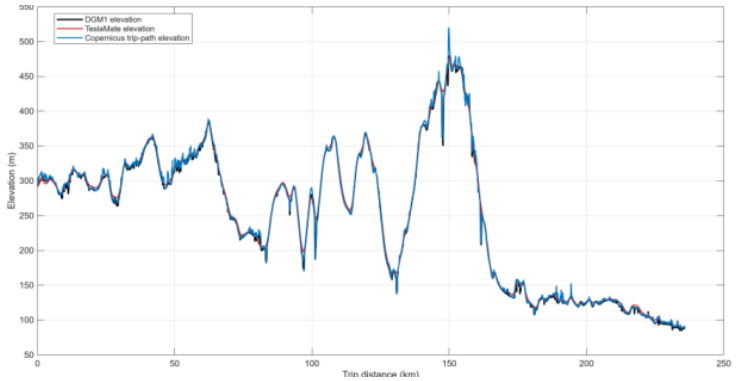
Input	Engineering role
Speed and acceleration	Inertial and aerodynamic power demand
Battery power	Measured reference for calibration and residual learning
Outside temperature	Context for unmodelled auxiliary and environmental effects
Elevation and route position	Terrain reconstruction; original coordinates not released

Data governance

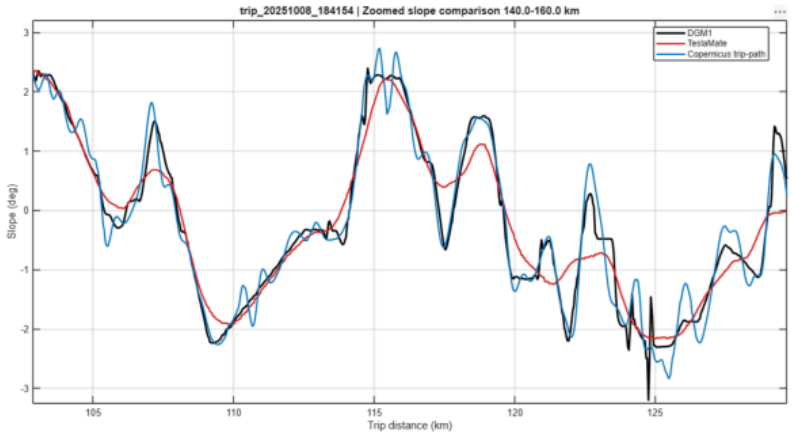
The public document reports methods and aggregated results only. It does not contain the original CSV files, route maps with identifiable locations, exact latitude/longitude records, timestamps linked to locations, or derived datasets from which trips could be reconstructed.

03 / ROAD-SLOPE RECONSTRUCTION

Road slope materially affects the gravitational component of longitudinal vehicle load. Directly differentiating noisy vehicle-derived elevation can create unrealistic slope spikes, so the project compared vehicle-side elevation with Copernicus GLO-30 and high-resolution DGM1 terrain data.



Source figure 4.7. Elevation-source comparison along a representative route; shown against distance rather than publishing a route map.



Source figure 4.8. Road-slope estimates obtained from alternative elevation sources.

DGM1-derived road slope was selected for the final model because it provided the most consistent terrain representation in the studied routes. This is a selection supported by the dataset comparison, not proof of universally superior performance across all road geometries.

04 / PHYSICS-BASED VEHICLE DYNAMICS

The baseline estimates longitudinal tractive force from inertia, rolling resistance, aerodynamic drag and road gradient. The wheel-power estimate is then translated to battery power using drivetrain and regenerative-efficiency assumptions together with auxiliary demand.

```
F_total = F_inertia + F_roll + F_aero + F_grade
```

```
P_wheel = F_total x vehicle speed
```

The force balance provides a physically interpretable foundation: aerodynamic load rises with the square of speed, while gradient load changes with road slope and vehicle mass. In practice, battery power also reflects conversion losses and accessory loads, which are not completely observed in the recorded signals.

Modelling boundaries

The project does not directly measure wind, tyre pressure, varying payload, detailed drivetrain operating points or thermal-system demand. Such effects may remain in the difference between measured and modelled battery power; this motivates the subsequent residual-learning stage.

05 / LEAST-SQUARES PARAMETER IDENTIFICATION

A least-squares calibration procedure identified a common parameter set for the physics-based baseline using measured battery-power information. Short analysis windows were used to form the estimation objective, with bounded parameters to maintain physically plausible ranges.

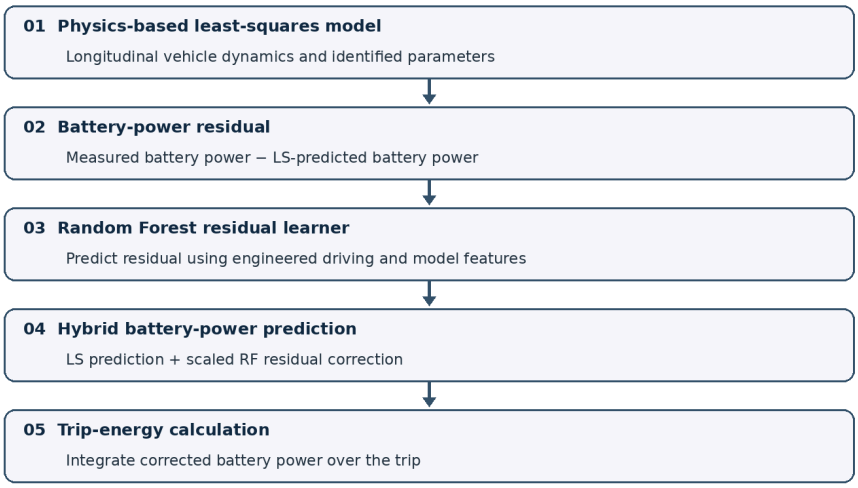
Calibrated quantity	Reported result
Aerodynamic drag area, CdA	0.4121 m²
Rolling resistance scale factor	1.1500
Drivetrain efficiency	0.9700
Regenerative efficiency	0.5000

Several identified parameters reached their optimization bounds. This matters: the estimates should be interpreted as effective calibration parameters for the chosen model and data, not as independently verified physical properties of the vehicle.

Why a physics-only model was insufficient

The baseline captured broad power and trip-energy trends but retained errors during transient and unobserved operating conditions. These residuals provided the target for a second, data-driven correction stage. The hybrid model was designed to supplement rather than replace the physical force balance.

06 / HYBRID RESIDUAL-LEARNING FRAMEWORK



Source figure 5.1. Original architecture of the physics-based baseline and Random Forest residual correction.

The Random Forest model learned the battery-power residual, defined as measured battery power minus the least-squares estimate. Twenty-five engineered features were used to represent driving and model context. The final hybrid estimate combined the physics prediction with a scaled predicted residual.

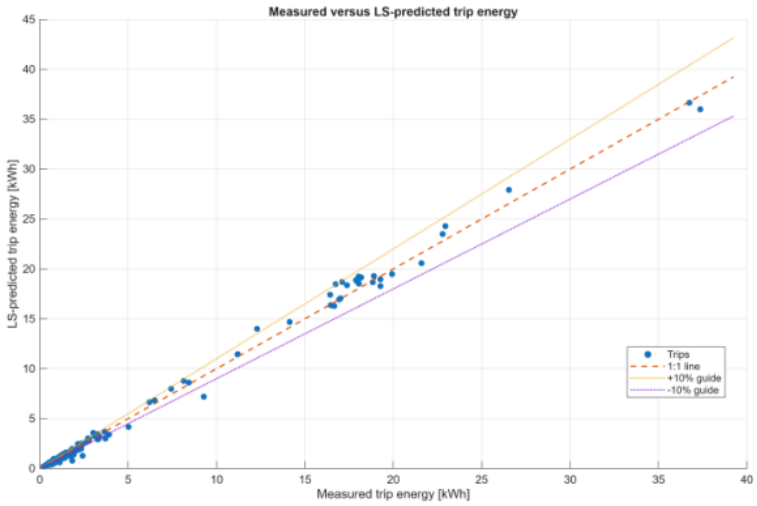
```
P_hybrid(t) = P_LS(t) + gamma x r_RF(t)
```

The residual correction strength was selected using validation data. The reported final configuration used **gamma = 0.65**, 300 trees, minimum leaf size 50 and 14 candidate predictors per split. These choices describe the reported experimental model, not an independently optimized production configuration.

07 / EVALUATION DESIGN

The **404 trips** were divided into training, validation and test subsets using an approximately **70/15/15 trip-level split**. The final reported test subset contained 62 trips. The model was assessed at both sample-wise battery-power and integrated trip-energy levels.

Metric	Purpose
Battery-power MAE and RMSE	Assess local prediction differences in kW
Mean absolute trip-energy percentage error	Assess integrated energy accuracy over trips
Improved-trip count	Count trips whose absolute energy error decreased



Source figure 6.2. Measured versus physics-model predicted trip energy across the recorded dataset.

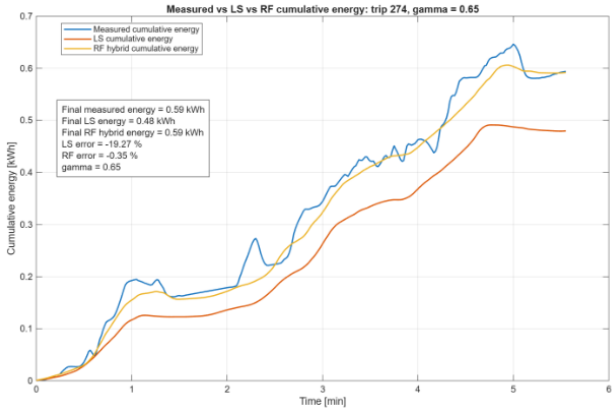
The primary public headline is the held-out test result. Aggregate results across all 404 trips are reported separately and should not be described as independent generalization performance.

08 / RESULTS AND ENGINEERING INTERPRETATION

Evaluation subset	LS error	Hybrid error	Relative reduction
Validation	12.68%	8.35%	34.2%
Test (62 trips)	12.59%	7.86%	37.6%
All 404 trips	10.85%	6.55%	39.6%

The hybrid model reduced mean absolute trip-energy percentage error on the test subset and improved **53 of 62 individual test trips**. **Nine test trips worsened.** The strongest absolute improvements occurred in shorter-duration groups, although long-trip groups contained very few examples.

model.



Source figure 6.12. Measured, least-squares and hybrid cumulative energy for one representative test trip.

For this illustrative trip, the absolute final-energy error decreased from **19.27% to 0.35%**. This example is not representative of every test trip; the aggregate test result above is the appropriate headline.

09 / LIMITATIONS, CONCLUSIONS AND RELEASE SCOPE

What the findings support

Within the recorded single-vehicle dataset, combining a calibrated longitudinal-dynamics baseline with Random Forest residual learning improved trip-energy estimates for most held-out test trips. The terrain comparison supported using DGM1-derived road slope as an input to the reported workflow.

Important limitations

The evaluation used one Tesla Model 3 and a limited set of driving environments. Performance on other vehicles, seasons and operating regimes was not established. Some fitted parameters reached their optimization bounds, indicating uncertainty or model compensation. Random Forest predictions may degrade when operating conditions differ from those seen during training. Reported trip-energy metrics should not be interpreted as physical battery-efficiency gains.

Potential next investigations

Useful future work includes cross-vehicle testing, broader weather and route coverage, explicit uncertainty analysis, stronger baselines and online or physics-constrained residual models. These are prospective directions, not completed validations.

Public repository contents

This portfolio edition is a narrative and results summary. The original report, MATLAB scripts, raw TeslaMate recordings, derived trip files and precise route information are intentionally excluded. Figures are adapted from the author's thesis report and should not be used to infer the release of the underlying dataset.

Source: Project thesis, “Vehicle Load Modeling Based on Real-World Driving Profiles and Dynamics,” Sections 1–7; source figures 4.7, 4.8, 5.1, 6.2 and 6.12.